

Equal consecutive Ulm quotients: a counterexample to 00000004273

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Statement and actual invariants

The conjecture describes the Ulm cardinal sequence of a countable p -group and requires realizable sequences to decrease strictly in ordinal order. We exhibit a finite abelian 2-group with two equal, nonzero consecutive Ulm invariants. Neither terminal zeroes nor an infinite-stage convention is involved.

Take the actual additive group $G = \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$. It is finite, hence countable, and $4x = 0$ for every $x \in G$; therefore it is 2-primary. Write $U_n = (2^n G)[2]$, the subgroup of elements annihilated by 2 inside the image of multiplication by 2^n . At each finite stage the corresponding Ulm quotient is $Q_n = U_n/U_{n+1}$. We use its actual cardinality; the usual vector-space-dimension convention gives the same equal adjacent values in this example.

Theorem 1. *The actual Ulm quotient cardinalities of this countable 2-group do not form a strictly decreasing sequence: $|Q_0| = |Q_1| = 2$.*

Proof. In the indicated coordinate groups,

$$\begin{aligned} U_0 &= \{(0, 0), (1, 0), (0, 2), (1, 2)\}, \\ U_1 &= \{(0, 0), (0, 2)\}, \quad U_2 = \{(0, 0)\}. \end{aligned}$$

The first-coordinate map $U_0 \rightarrow \mathbb{Z}/2\mathbb{Z}$ is a surjective homomorphism with kernel U_1 . The map $U_1 \rightarrow \mathbb{Z}/2\mathbb{Z}$ sending $(0, 0)$ to 0 and $(0, 2)$ to 1 is a surjective homomorphism with kernel U_2 . The actual quotient groups Q_0 and Q_1 are therefore both isomorphic to $\mathbb{Z}/2\mathbb{Z}$, by the first isomorphism theorem. Their cardinals are equal and nonzero. Strict anti-monotonicity would require $|Q_1| < |Q_0|$, a contradiction. If Ulm entries are instead recorded as dimensions over $\mathbb{F}_2$, these same quotients are both one-dimensional, so strict decrease still fails. The formal capstone uses literal cardinal-valued entries. $\square$

An ordinal-indexed strictly decreasing sequence has strictly decreasing restriction to its indices 0 and 1. Failure on this finite pair thus refutes the proposed necessary condition for realizability, regardless of how later ordinal stages or a “full” sequence are presented. The additional lattice-closure clause is not needed.

Formal verification and earlier submission

Lean uses Mathlib’s actual additive group `ZMod 2 × ZMod 4`, homomorphism images and kernels, additive subgroups, and actual quotient groups. It verifies the subgroup descriptions, the two surjective coordinate homomorphisms, their kernels, quotient equivalences, and both natural and literal `Cardinal.mk` cardinalities. The final theorem simultaneously proves countability, the 2-primary condition, and failure of standard `StrictAnti` for the actual cardinal sequence.

Earlier closed PR #309 used this same finite counterexample, but its capstone encoded the supposed strict decrease as a tailored prohibition on two quotient certificates rather than directly negating strict decrease of a cardinal-valued sequence. This replacement supplies that semantic bridge, uses Mathlib’s quotient objects, and credits the prior example; it does not claim that PR #309 received a rejecting review. The development compiles on Lean 4.33.1 with pinned Mathlib. It has no unfinished proofs, native evaluation, or extra axioms. The only printed dependencies are propositional extensionality, classical choice, and quotient soundness. The tiny coordinate checks range over eight group elements, not a long brute-force computation.